NutriVision

AI-Driven Dietary Health Assistant Application

Person 1 Deliverables | System Analysis & Design

1. Problem Statement & Objectives

Problem Statement

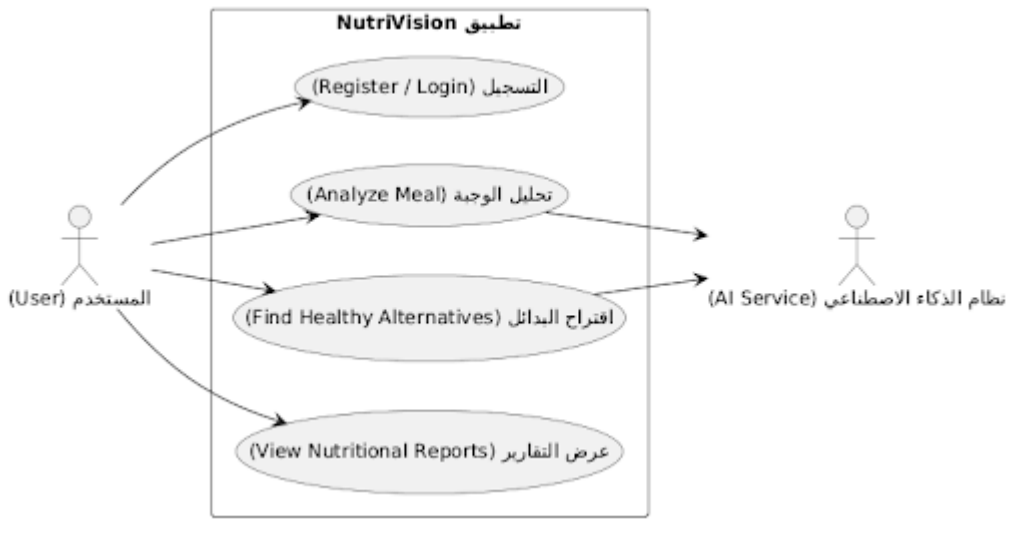
Individuals seeking a healthy lifestyle struggle to accurately track and analyze the nutritional value of their daily meals. Furthermore, they lack access to personalized, healthy alternatives that accommodate their dietary restrictions without disrupting their daily routines.

Project Objectives:

- Provide accurate and fast meal analysis utilizing AI and natural language processing.
- Offer personalized healthy food and snack alternatives based on individual dietary constraints and preferences.
- Enable real-time, interactive nutritional reports to track and maintain dietary logs across multiple devices.

2. Use Cases & Diagram

Use Case	Actor	Description & Functional Importance
Register / Login	User	Create a new account or log in to securely save personal data and cloud-based dietary logs.
Analyze Meal	User + AI Service	Input meal descriptions for the AI to analyze ingredients and generate nutritional insights and values.
Find Alternatives	User + AI Service	Search for healthy alternatives to specific snacks and meals based on user preferences.
Nutritional Reports	User	View interactive charts and daily/weekly meal logs generated from collected data.



Approved Use Case Diagram for NutriVision

3. Architecture Pattern & Implementation

The **Clean Architecture with MVVM (Model-View-ViewModel)** pattern was chosen to ensure a complete separation of business logic from the user interface and data sources, facilitating easier testing and future scalability.

1. Presentation Layer

- Responsive UI utilizing `GridView` and `Stack`.
- Displays meal logs and charts.
- State management via ViewModels/Bloc.

2. Domain Layer

- Contains independent Use Cases.
- `AnalyzeMealUseCase`
- `GetAlternativesUseCase`
- Free of external dependencies.

3. Data Layer

- Utilizes `Dio` with Interceptors for AI API requests.
- Integrates `Firebase Firestore` for real-time log synchronization.